

Exercise 7 – Routing, Route Guards and Lazy Loading

Generate Components and Guards

```
ng generate component pages/login
ng generate component pages/dashboard
ng generate component pages/not-found
ng generate guard guards/auth
ng generate guard guards/unsaved-changes
```

auth.service.ts

```
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root'
})
export class AuthService {
  private loggedIn = false;

  login() {
    this.loggedIn = true;
  }

  logout() {
    this.loggedIn = false;
  }

  isLoggedIn() {
    return this.loggedIn;
  }
}
```

auth.guard.ts

```
import { inject } from '@angular/core';
import { CanActivateFn, Router } from '@angular/router';
import { AuthService } from '../services/auth.service';
export const authGuard:CanActivateFn={()=>{
const auth=inject(AuthService);
const router=inject(Router);
if(auth.isLoggedIn()){
return true;
}
router.navigate(['/login']);
return false;
};
```

unsaved-changes.guard.ts

```
import { CanDeactivateFn } from '@angular/router';
import { EnrollmentFormComponent } from '../pages/enrollment-form/enrollment-form.component';
export const unsavedChangesGuard:CanDeactivateFn<EnrollmentFormComponent>=(
component
)>=>{
return confirm('Discard Unsaved Changes?');
};
```

login.component.ts

```
import { Component } from '@angular/core';
import { Router } from '@angular/router';
import { AuthService } from '../services/auth.service';
@Component({
```

```
selector:'app-login',
standalone:true,
templateUrl: './login.component.html',
styleUrls: ['./login.component.css']
})

export class LoginComponent{
username="";
password="";
constructor(
private auth:AuthService,
private router:Router
){}
login(){
if(this.username==='admin'&&this.password==='admin'){
this.auth.login();
this.router.navigate(['/dashboard']);
}else{
alert('Invalid Credentials');
}
}
}
```

login.component.html

```
<h2>Login</h2>

<input
type="text"
[(ngModel)]="username"
placeholder="Username">
```

```
<input
type="password"
[(ngModel)]="password"
placeholder="Password">
<button (click)="login()">
Login
</button>
```

dashboard.component.ts

```
import { Component } from '@angular/core';
@Component({
  selector:'app-dashboard',
  standalone:true,
  templateUrl:'./dashboard.component.html',
  styleUrls:['./dashboard.component.css']
})
export class DashboardComponent{}
```

dashboard.component.html

```
<h2>Student Dashboard</h2>
<p>
Welcome to Angular Student Portal
</p>
```

not-found.component.ts

```
import { Component } from '@angular/core';
@Component({
  selector:'app-not-found',
```

```
standalone:true,  
templateUrl: './not-found.component.html'  
})  
export class NotFoundComponent{
```

not-found.component.html

```
<h2>404</h2>  
  
<p>  
Page Not Found  
</p>
```

app.routes.ts

```
import { Routes } from '@angular/router';  
  
import { HomeComponent } from './pages/home/home.component';  
  
import { CourseListComponent } from './pages/course-list/course-list.component';  
  
import { StudentProfileComponent } from './pages/student-profile/student-profile.component';  
  
import { EnrollmentFormComponent } from './pages/enrollment-form/enrollment-form.component';  
  
import { ReactiveEnrollmentComponent } from './pages/reactive-enrollment/reactive-enrollment.component';  
  
import { LoginComponent } from './pages/login/login.component';  
  
import { DashboardComponent } from './pages/dashboard/dashboard.component';  
  
import { NotFoundComponent } from './pages/not-found/not-found.component';  
  
import { authGuard } from './guards/auth.guard';  
  
import { unsavedChangesGuard } from './guards/unsaved-changes.guard';  
  
export const routes:Routes=[  
{path:"",component:HomeComponent},
```

```
{path:'courses',component:CourseListComponent},
{path:'profile',component:StudentProfileComponent},
{path:'enroll',component:EnrollmentFormComponent,canDeactivate:[unsavedChangesGuard]},
{path:'reactive',component:ReactiveEnrollmentComponent},
{path:'login',component:LoginComponent},
{path:'dashboard',component:DashboardComponent,canActivate:[authGuard]},
{path:'**',component:NotFoundComponent}
];
```

header.component.html

```
<header>
<h1>{{title}}</h1>
<nav>
<a routerLink="/">Home</a>
<a routerLink="/courses">Courses</a>
<a routerLink="/profile">Profile</a>
<a routerLink="/enroll">Template Form</a>
<a routerLink="/reactive">Reactive Form</a>
<a routerLink="/login">Login</a>
<a routerLink="/dashboard">Dashboard</a>
</nav>

</header>
```

app.component.html

```
<app-header></app-header>
```

```
<router-outlet></router-outlet>
```

Run

ng serve

Open

http://localhost:4200

Output Screenshot:

